



**European
University Cyprus**

Assignment 3 : Credit Card Default Prediction

Course: AI 605 Introduction to Machine Learning

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1. Goal

The goal of this assignment is to work on an important problem for banks, which is to predict and identify clients who are at high risk of not paying their credit card payments. To do this, different machine learning classification models are developed and compared.

The main objective is to evaluate the performance of these models and decide which classification model gives the best results for predicting credit default

2.Dataset

The dataset used in this assignment is an Excel file that contains information about 30,000 credit card clients. Each row in the dataset represents one client, and the dataset includes 23 input features along with one target variable.

The features describe demographic information, credit information, and monthly repayment behavior. Some features represent the same type of information measured over different months. These features are grouped as follows:

- **Demographic features:**
Gender, education level, marital status, and age.
- **Credit information:**
Credit limit assigned to the client.
- **Monthly repayment status:**
Repayment status for the past several months, indicating delays or timely payments.
- **Monthly bill amounts:**
Bill statement amounts recorded for multiple consecutive months.
- **Monthly payment amounts:**
Amounts paid by the client in each of the recorded months.

The target variable indicates whether a client defaulted on their credit card payment, where 1 represents a defaulter and 0 represents a non-defaulter.

The dataset was obtained from the UCI Machine Learning Repository, which provides publicly available datasets for research and educational purposes. The dataset can be found at: <https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients>

3. Approach

1. Check the dataset and review the available features.
2. Decide whether any features should be removed (all features were kept).

3. Choose suitable classification models to work with the data.
4. Split the dataset into training and testing sets using a fixed percentage.
5. Select evaluation metrics to properly evaluate the models.
6. Implement the models in code and analyze the results.

4. Reasoning and Decisions

- **Dataset:**

The dataset was not changed because there are no unnecessary features. It also has a good variety of input data, so all features were kept.

- **Classification Models:**

Logistic Regression was chosen because it is a simple model. It was used to see how it performs with many input features and to have a basic model for comparison.

Decision Tree was chosen because it works well with many features and different types of data.

Random Forest was chosen as the main model because it is made of many decision trees and can handle many features better.

- **Train/Test Split:**

A 70/30 split was chosen so that there is enough data for training and also enough data for testing the models.

- **Evaluation Metrics:**

In this problem, it is important to reduce the number of defaulters that are predicted as non-defaulters. For this reason, recall and F1-score were used. Precision and confusion matrices were also checked to see the errors clearly. ROC-AUC was used to see how stable and reliable the models are.

5. Results

The results of the three models are shown in a summary table with accuracy, balanced accuracy, recall, F1-score, and ROC-AUC values.

Logistic Regression gives a relatively high accuracy, but it almost does not detect defaulters. This can be seen from the very low recall for the default class, which makes the model not suitable for this problem.

The Decision Tree model performs better than Logistic Regression and is able to detect more defaulters. However, it also makes more mistakes by predicting some non-defaulters as defaulters.

The Random Forest model shows the best overall results. It has the highest ROC-AUC and F1-score, and it gives a better balance between detecting defaulters and avoiding too many false predictions.

Confusion matrices were used to visually check how many clients were classified correctly and incorrectly by each model.

Model	Accuracy	Balanced Accuracy	Recall (Defaulters)	F1- Score	ROC-AUC
Logistic Regression	0.78	0.5	0.00	0.00	0.65
Decision Tree	0.72	0.61	0.41	0.39	0.69
Random Forest	0.76	0.65	0.36	0.46	0.76

6. Comparison and Discussion

By comparing the results of the three models, clear differences can be observed. Logistic Regression performed the worst for this problem. Although it achieved relatively high accuracy, it almost failed to identify defaulters, which is a serious issue for a banking application. This shows that accuracy alone is not a good metric when the dataset is imbalanced.

The Decision Tree model performed better than Logistic Regression. It was able to detect more defaulters, but it also produced more false positives by incorrectly classifying some non-defaulters as defaulters. This shows a trade-off between catching risky clients and making wrong predictions for safe clients.

The Random Forest model performed the best overall. It achieved the highest ROC-AUC and F1-score, which means it can better separate defaulters from non-defaulters while keeping a reasonable balance between errors. For this reason, Random Forest is the most suitable model among the ones tested.

One main difficulty of the dataset is class imbalance, since non-defaulters are much more frequent than defaulters. This makes the problem harder and explains why some models tend to predict most clients as non-defaulters. Another challenge is the presence of many monthly features, which can be difficult for simpler models to handle.

Overall, the comparison shows that more complex models, such as Random Forest, are better suited for this type of data, while simpler models may fail to capture important patterns related to credit default.

7. Conclusion

In this assignment, different machine learning models were tested to predict credit card default. The results show that this is not an easy problem, mainly because the dataset is imbalanced and defaulters are harder to detect.

Random Forest performed better than Logistic Regression and Decision Tree, but overall none of the models gave very strong results. Many defaulters were still missed, which shows the limitations of the models used.

This assignment shows that accuracy is not enough for this type of problem and that other metrics, such as recall and F1-score, are more important.

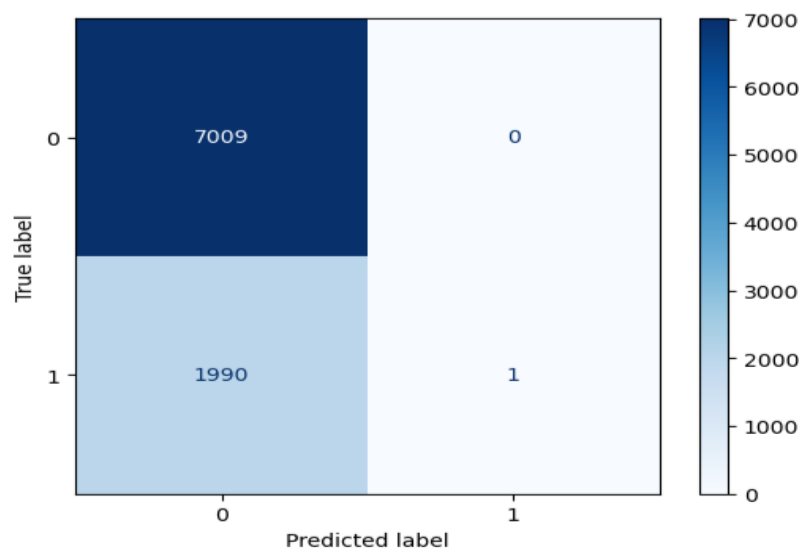
8. Reference :

- I used ChatGPT and Grammarly for better writing skills by putting my unclear and confusing sentences to improve them due to my Dyslexia. I tried mostly with my own writing skills.
- Anaconda Assistant for some errors in my code, for example, to read the Excel file
- UCI DATASET: <https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients>

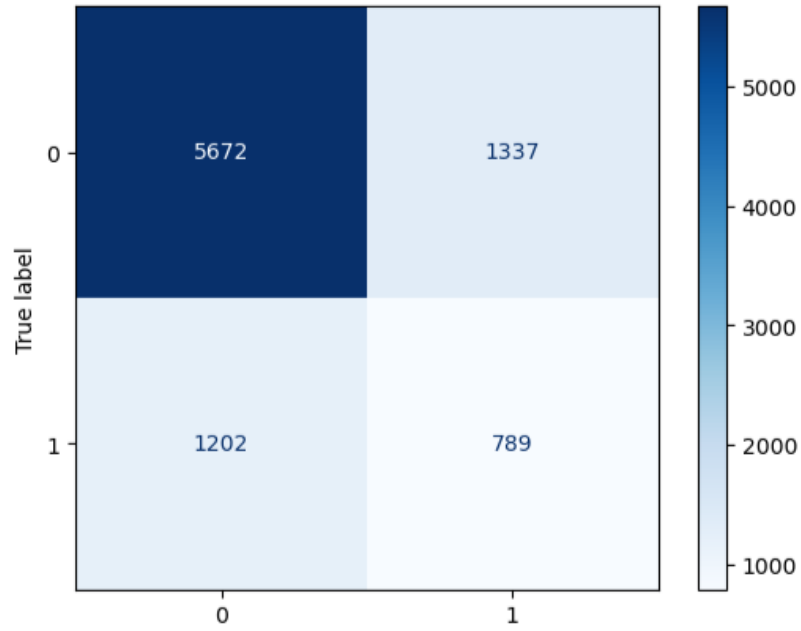
9. Appendix

Confusion Matrices for each model

Logistic Regression



Decision Tree



Random Forest

